You will be teaching a **30-minute** lesson in CSC 202: Data Structures and Object-Oriented Programming on Wednesday, February 25.

CSC 202 is the second course of our introductory sequence. The first course is taught in Python and required by students pursuing a computer science major or minor, a data major or minor, and an applied math major. A variety of other students (GIS, Engineering, Chemistry, etc.) that feel an introductory programming course would be important for their future goals also take CSC 201. Most CSC 202 students are pursuing a computer science major or minor since CSC 202 is required for both.

CSC 202 meets 75 minutes MWF for "lecture" and 150 minutes on Tuesday for a lab. The first 6 weeks of CSC 202 transitions students to Java (basic control structures, arrays, ArrayLists) and focuses on object-oriented programming (user-defined classes, inheritance, and interfaces). Beginning week 7, we shift our focus to data structures and their implementations.

Your lesson is our 10th MWF class session of the term. Prior to your lesson, we will have "covered" input from the keyboard using a Scanner; output to the monitor using System.out.print, println, and printf; primitive data types; assignment statements; conditional statements and loops; methods with parameters both returning and not returning a value; using and defining a user-defined ("custom") class; 1D and 2D arrays; methods of the String class; and methods of the Random class. I've included a calendar of topics for the course so that you can also see what topics we have yet to "cover".

CSC 202 is taught in a classroom with an instructor station including a desktop computer (with PowerPoint and IntelliJ installed), a white board, a projector, and a document camera. Your laptop can also be connected to the projector via HDMI (or with a USB-C to HDMI adapter). The class has 22 students, generally sitting at tables facing the front of the room. Most students will have their laptops with them. If you think it would be helpful, you are welcome to ask them to use their laptops during your lesson. Or if you would prefer they put their laptops away and focus attention on your lesson, that's fine too. The students used IntelliJ IDE (Community Edition) for the lab on Tuesday, Feb 10 but not all have it working on their laptops yet.

Your lesson topic is ArrayLists.

The objectives for your 30-minute lesson are:

1) Briefly introduce ArrayLists and the basic methods of the ArrayList class (i.e. add(value), add(index, value), size(), get(value), indexOf(value), set(index, value), remove(index), remove(value), clear(). *(Note: the students have prior experience using **lists** in Python, so these operations on a dynamic-length list should be familiar, although the syntax differs.)*

2) Briefly compare/contrast ArrayLists and 1D arrays.

3) Write code for an application using an ArrayList of some custom data type (not a built-in class like Strings or Integers) to do something useful/interesting.

	Monday	Tuesday	Wednesday	Friday
1			**February 4** Java Basics	**February 6** Conditional Execution
2	**February 9** Loops	**February 10** LAB 1	**February 11** User Defined Classes	**February 13** User Defined Classes
3	**February 16** Class Composition & Static Variables/Methods	**February 17** LAB 2	**February 18** 1D Arrays	**February 20** 2D Arrays
4	**February 23** String Manipulations/ Random Numbers	**February 24** LAB 3	**February 25** ArrayList	**February 27** Exceptions
5	**March 2** Exceptions & File I/O	**March 3** LAB 4	**March 4** Inheritance	**March 6** Inheritance & Abstract Classes
6	**March 9** Review & Mini-Lab 5	**March 10** **Midterm 1**	**March 11** Polymorphism & Interfaces	**March 13** Equals & Comparable Interface
7	**March 16** Efficiency & Big O	**March 17** LAB 6	**March 18** ArrayIntList	**March 20** Stack & Queue
			Spring Break Week	
8	**March 30** Writing Code with Stack & Queue	**March 31** LAB 7	**April 1** List Nodes	**April 3** **No Class—Easter Break**
9	**April 6** **No Class—Easter Break**	**April 7** LAB 8	**April 8** Singly Linked List	**April 10** Singly Linked List cont.
10	**April 13** Linked List Varieties	**April 14** LAB 9	**April 15** Recursive Tracing	**April 17** Writing Recursive Methods
11	**April 20** Review	**April 21** **Midterm 2**	**April 22** Applications of Recursion	**April 24** Sets & Maps
12	**April 27** Writing Methods using Sets & Maps	**April 28** LAB 10	**April 29** HashSet & HashMap Implementation	**May 1** Binary Tree
13	**May 4** Writing Binary Tree Methods	**May 5** **Coding Exam**	**May 6** **Celebration of Learning**	**May 8** Complete Binary Tree/ Begin Binary Search Tree
14	**May 11** Binary Search Tree—Writing Methods	**May 12** LAB 11	**May 13** Binary Search Tree—Removing Nodes	**May 15** Wrap up Binary Trees & Review